

Solar Powered Iron Press System - Thermal Calculations

This document contains basic engineering calculations related to the Solar Powered Iron Press System using solar thermal concentration.

Parameter	Value
Solar Intensity (I)	1000 W/m ²
Fresnel Lens Area (A)	0.25 m ²
Estimated Thermal Efficiency	72–80%

1. Heat Energy Calculation

Formula:

$$Q = I \times A$$

Where:

Q = Heat Energy Collected

I = Solar Intensity

A = Fresnel Lens Area

Substituting values:

$$Q = 1000 \times 0.25$$

$$Q = 250 \text{ W}$$

2. Useful Heat After Losses

Assuming thermal losses due to convection and radiation:

Useful Heat $\approx$ 72% to 80% of total collected heat

Useful Heat $\approx$ 180–200 W

3. Thermal Efficiency Estimation

Thermal Efficiency Formula:

$$\eta = (\text{Useful Heat Output} / \text{Total Solar Heat Input}) \times 100$$

Example:

$$\eta = (190 / 250) \times 100$$

$$\eta \approx 76\%$$

These calculations provide an estimated thermal performance of the Solar Powered Iron Press System and demonstrate the feasibility of solar thermal energy for domestic ironing applications.